



EV Charging Behavior: Analysis Analysis and Prediction

Understanding EV charging is key for infrastructure planning, user modeling, and cost efficiency as electric vehicles gain traction. This project analyzes a simulated EV charging dataset to identify patterns and predict charging costs and energy demand, addressing complexities from growing charging networks and diverse vehicle models.

Data Cleaning and Transformation

The initial step involves preparing the dataset for analysis. This includes importing necessary libraries, loading the CSV file, and inspecting the data types of each column. Date/time columns are converted to datetime objects, and numerical columns are verified and converted as needed. Missing values are handled by dropping rows with incomplete data.

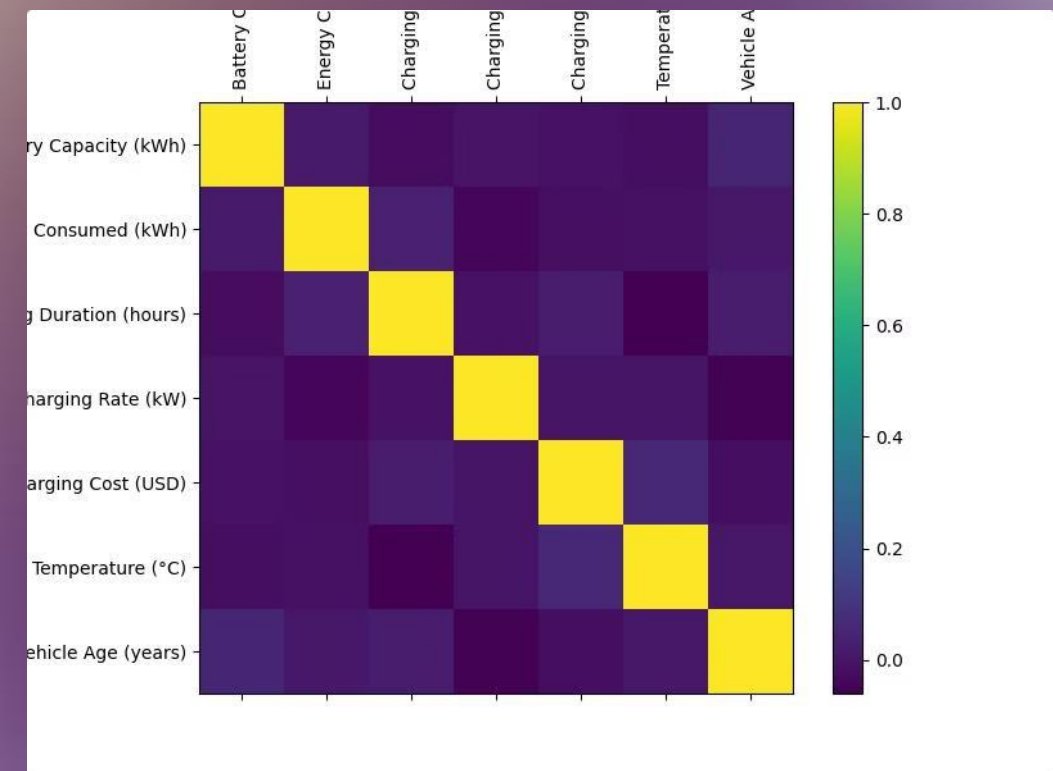
- Import libraries: **pandas**, **numpy**, **datetime**
- Load the dataset from CSV file
- Convert date columns to datetime objects, replacing invalid entries with **NaT**
- Ensure numeric columns are correctly formatted, converting invalid entries to **NaN**
- Handle missing values by dropping rows with **NaN** values

Data Visualization: Correlation

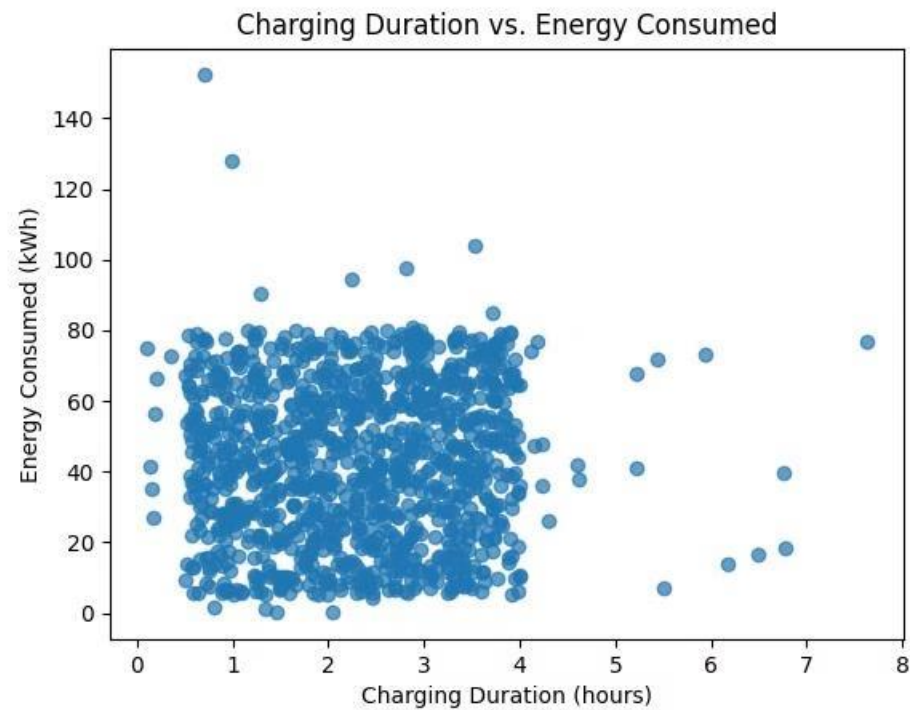
Correlation Matrix

A correlation matrix visualizes linear relationships between numeric variables. The correlation coefficient ranges from +1 (perfect positive) to -1 (perfect negative), with 0 indicating no correlation.

- Diagonal: Perfect self-correlation (value of 1).
- Off-diagonal: Weak linear relationships (values near 0).
- Positive Correlation: **Charging Duration & Energy Consumed**, and **Charging Cost & Energy Consumed**.
- No Strong Multicollinearity: Reduces data redundancy.



Data Visualization: Scatter Plot



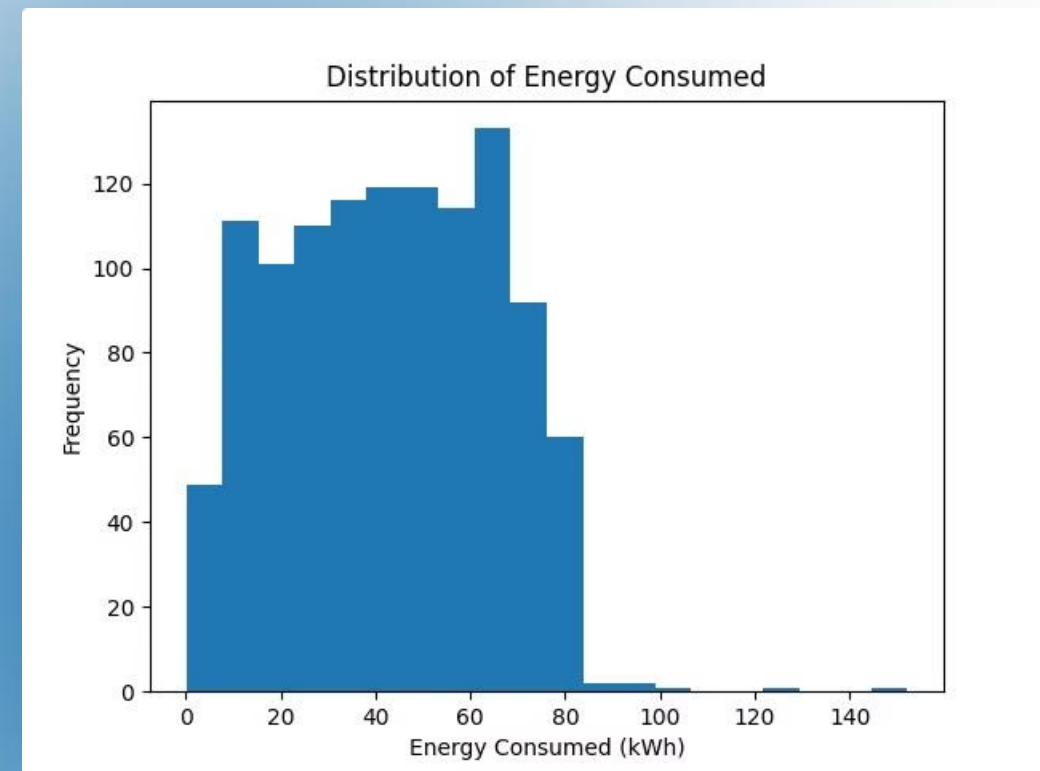
A scatter plot of Charging Duration vs. Energy Consumed highlights potential relationships. Expect longer sessions to use more energy, influenced by charger type and battery size.

- Upward trend suggests a positive correlation: longer charging, more energy consumed.
- Data spread indicates time isn't the only factor determining energy use.
- Outliers represent potentially inefficient charging or unique cases.

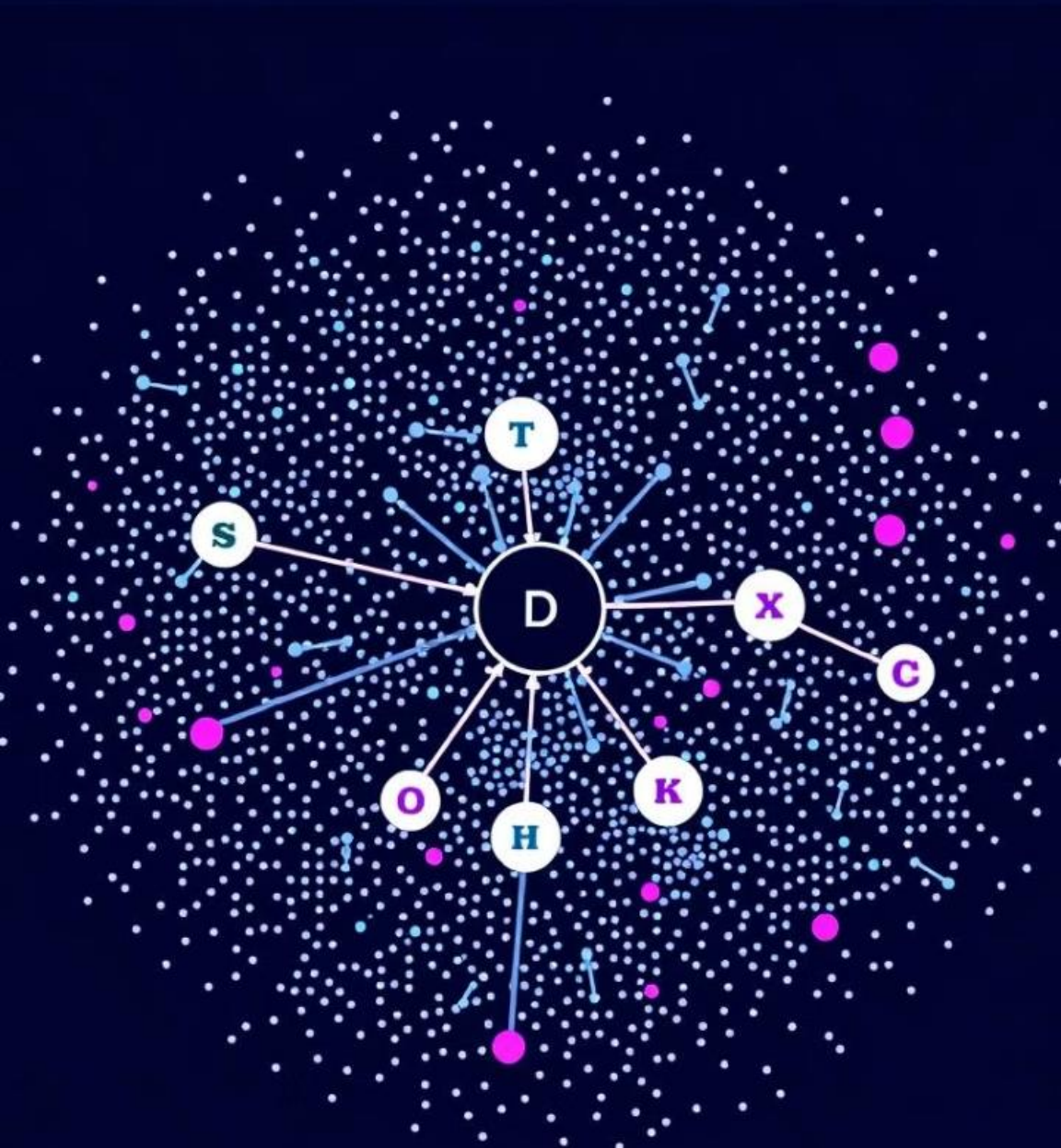
Data Visualization: Histogram of Energy Consumed

This histogram illustrates the distribution of energy consumption (kWh) during EV charging sessions, highlighting common usage ranges and outliers.

- Most sessions consume between 20 kWh and 70 kWh.
- The distribution is right-skewed, indicating moderate energy use is frequent.
- Outliers show very high energy consumption (>100 kWh), possibly due to long trips or large battery EVs using fast chargers.
- Analyzing demand helps optimize infrastructure planning.



KMeans Clustting



Data Labeling: K-Means Clustering

K-Means Clustering categorizes charging sessions based on behavior. A new "Cluster Label" column groups sessions by Charging Duration, Energy Consumed, Charging Rate, and Charging Cost for further analysis.

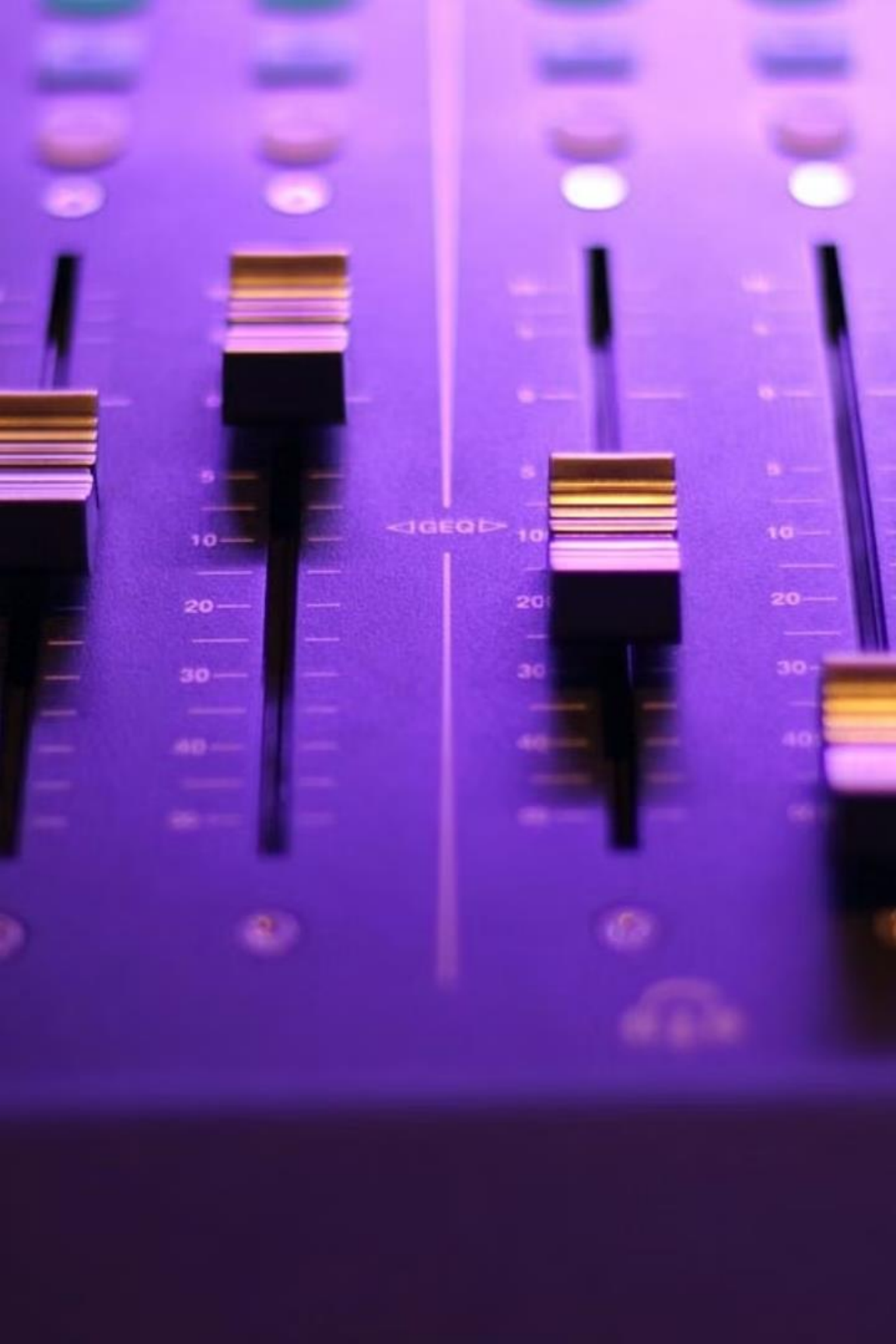
- Cluster 0: Most sessions (419), likely medium energy, average cost, moderate duration.
- Cluster 1: 387 sessions, possibly short, fast charges (DC fast chargers).
- Cluster 2: Fewest sessions (325), potentially long, low-rate charges (home/slow stations).



Model for Cost Prediction: Random Forest

A Random Forest Regressor predicts "Charging Cost (USD)" using session features and environmental variables. Random Forests effectively handle non-linear relationships for accurate cost prediction. Model trained on session features.

- Initial Results: MSE: 41.90, R²: 0.66
- MSE: Average prediction error of $\pm \$6.47$.
- R²: Model explains 66% of charging cost variation.



Model Improvement: Hyperparameter Tuning with GridSearchCV

GridSearchCV was used to fine-tune the Random Forest Regressor, resulting in the best configuration: `max_depth=10`, `min_samples_leaf=2`, `min_samples_split=2`, and `n_estimators=200`. These parameters optimize tree structure and minimize overfitting.

- Expanded hyperparameters: Tests various `n_estimators`, `max_depth`, and `min_samples` settings.
- `cv=3` cross-validation: Validates model on different data subsets to prevent overfitting.
- Improved Results: MSE: 40.92, R2 Score: 0.67

Model for Charging Demand Prediction: Linear Regression

For predicting "Charging Demand" (Energy Consumed (KWh)), a Linear Regression model is initially used. Linear Regression provides an interpretable baseline for demand prediction, but the performance was poor ($R^2 \approx 0.39$). A Random Forest also underperformed ($R^2 \approx 0.31$).

- Linear Regression Results: MSE: 318.09, R^2 Score: 0.39
- Random Forest Results: MSE: 358.56, R^2 Score: 0.31
- The Linear Regression model's predictions are off by about ± 17.83 kWh on average.
- The Linear Regression model only explains 39% of the variation in energy consumption.



Results and Conclusion

K-Means clustering effectively segmented charging sessions into three distinct behavioral groups. Cluster 0 (419 sessions) represents medium energy consumption with average cost and moderate duration, while Cluster 1 (387 sessions) likely corresponds to short, fast charges from DC fast chargers, and Cluster 2 (325 sessions) potentially indicates long, low-rate charges from home or slow stations. The optimized Random Forest model, fine-tuned using GridSearchCV (max_depth=10, min_samples_leaf=2, min_samples_split=2, n_estimators=200), achieved an R^2 score of 0.67, demonstrating a meaningful ability to predict charging costs with a relatively low average prediction error of approximately $\pm\$6.47$.

- Cost Prediction: R^2 Score: 0.67 (Random Forest with GridSearchCV)
- Charging Demand Prediction: R^2 : 0.39 (Linear Regression)
- The Linear Regression model's lower performance in predicting charging demand ($R^2 = 0.39$) may stem from the inherent non-linear relationships between energy consumption and various session features. Future efforts should explore non-linear models or feature engineering to improve demand prediction accuracy.